MODULE - 3

(Infosys Springboard Virtual Internship 6.0)

SUBMITTED BY - ASHISH CHAUHAN

FILE STRUCTURE :

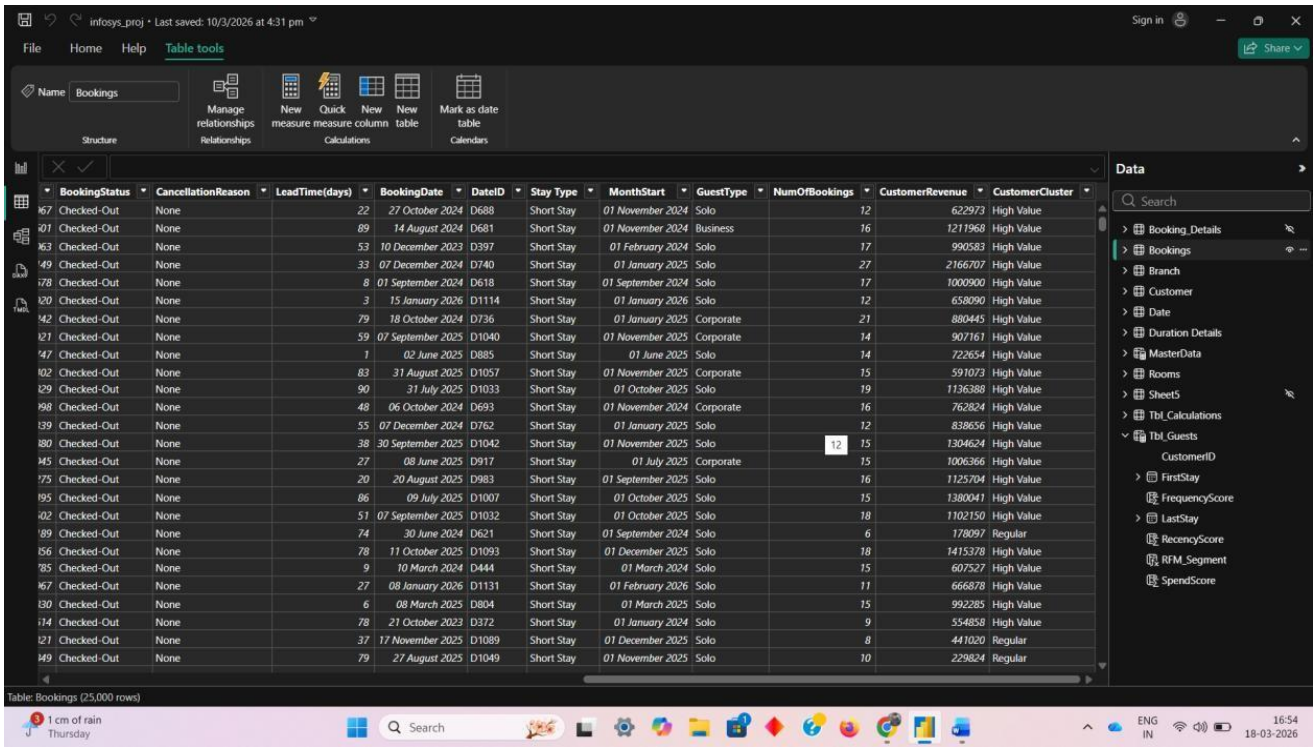
- Objective
- Dataset
- Segmentation
- Transformation
- RFM
- RFM Segmentation
- Insights

1. Objective

The objective of this module is to analyze guest behavior using segmentation techniques and RFM (Recency, Frequency, Monetary) analysis to identify valuable customers and improve business decision making.

2. Dataset Description

The dataset consists of booking-level data from a hotel management system. It includes key attributes such as CustomerID, BookingID, Checkin and Check-out dates, TotalAmount (revenue), Purpose of visit, Stay Type, and Booking Channel.
This data is used to analyze customer behavior and booking patterns.



BookingStatus	CancellationReason	LeadTime(days)	BookingDate	DateID	Stay Type	MonthStart	GuestType	NumOfBookings	CustomerRevenue	CustomerCluster
Checked-Out	None	22	27 October 2024	D688	Short Stay	01 November 2024	Solo	12	622973	High Value
Checked-Out	None	89	14 August 2024	D681	Short Stay	01 November 2024	Business	16	1211968	High Value
Checked-Out	None	53	10 December 2023	D397	Short Stay	01 February 2024	Solo	17	990583	High Value
Checked-Out	None	33	07 December 2024	D740	Short Stay	01 January 2025	Solo	27	2166707	High Value
Checked-Out	None	8	01 September 2024	D618	Short Stay	01 September 2024	Solo	17	1000900	High Value
Checked-Out	None	3	15 January 2026	D1114	Short Stay	01 January 2026	Solo	12	658090	High Value
Checked-Out	None	79	18 October 2024	D736	Short Stay	01 January 2025	Corporate	21	880445	High Value
Checked-Out	None	59	07 September 2025	D1040	Short Stay	01 November 2025	Corporate	14	907161	High Value
Checked-Out	None	1	02 June 2025	D885	Short Stay	01 June 2025	Solo	14	722654	High Value
Checked-Out	None	83	31 August 2025	D1057	Short Stay	01 November 2025	Corporate	15	591073	High Value
Checked-Out	None	90	31 July 2025	D1033	Short Stay	01 October 2025	Solo	19	1136388	High Value
Checked-Out	None	48	06 October 2024	D693	Short Stay	01 November 2024	Corporate	16	762824	High Value
Checked-Out	None	55	07 December 2024	D762	Short Stay	01 January 2025	Solo	12	838656	High Value
Checked-Out	None	38	30 September 2025	D1042	Short Stay	01 November 2025	Solo	15	1304624	High Value
Checked-Out	None	27	08 June 2025	D917	Short Stay	01 July 2025	Corporate	15	1006366	High Value
Checked-Out	None	20	20 August 2025	D983	Short Stay	01 September 2025	Solo	16	1125704	High Value
Checked-Out	None	86	09 July 2025	D1007	Short Stay	01 October 2025	Solo	15	1380041	High Value
Checked-Out	None	51	07 September 2025	D1032	Short Stay	01 October 2025	Solo	18	1102150	High Value
Checked-Out	None	74	30 June 2024	D621	Short Stay	01 September 2024	Solo	6	178097	Regular
Checked-Out	None	78	11 October 2025	D1093	Short Stay	01 December 2025	Solo	18	1415378	High Value
Checked-Out	None	9	10 March 2024	D444	Short Stay	01 March 2024	Solo	15	607527	High Value
Checked-Out	None	27	08 January 2026	D1131	Short Stay	01 February 2026	Solo	11	666878	High Value
Checked-Out	None	6	08 March 2025	D804	Short Stay	01 March 2025	Solo	15	992285	High Value
Checked-Out	None	78	21 October 2023	D372	Short Stay	01 January 2024	Solo	9	554858	High Value
Checked-Out	None	37	17 November 2025	D1089	Short Stay	01 December 2025	Solo	8	441020	Regular
Checked-Out	None	79	27 August 2025	D1049	Short Stay	01 November 2025	Solo	10	229824	Regular

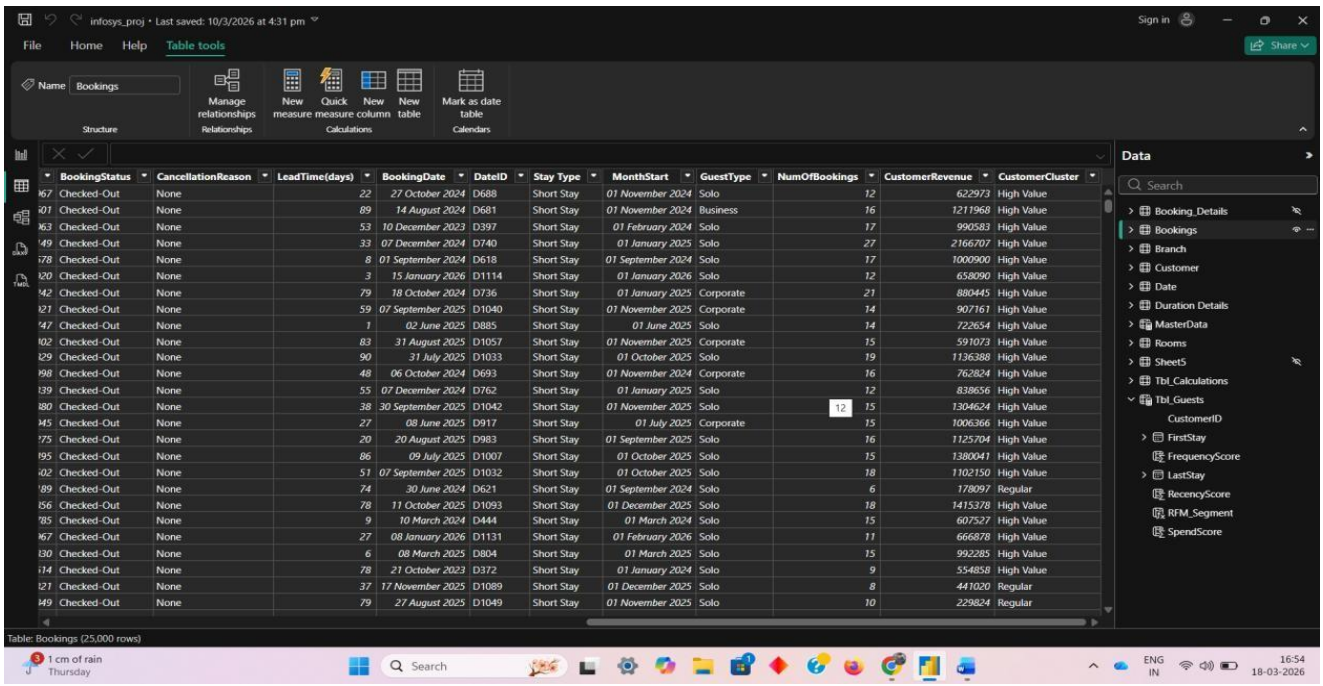
3. Guest Segmentation

1. GuestType

A calculated column "GuestType" was created using conditional logic, DAX functions such as SWITCH and CALCULATE to dynamically classify customers based on business logic such as Business, Family, Corporate, and Solo based on Purpose, Booking Channel, and Stay Type.

2. CustomerCluster

Customer segmentation was performed using booking frequency and total revenue. Customers were categorized into First-Timer, Regular, and High Value segments based on their booking behavior and spending patterns.



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63 Checked-Out	None	53	10 December 2023	D397	Short Stay	01 February 2024	Solo	17	990583	High Value
49 Checked-Out	None	33	07 December 2024	D740	Short Stay	01 January 2025	Solo	27	2166707	High Value
78 Checked-Out	None	8	01 September 2024	D618	Short Stay	01 September 2024	Solo	17	1000900	High Value
20 Checked-Out	None	3	15 January 2026	D1114	Short Stay	01 January 2026	Solo	12	658090	High Value
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21 Checked-Out	None	59	07 September 2025	D1040	Short Stay	01 November 2025	Corporate	14	907161	High Value
47 Checked-Out	None	1	02 June 2025	D885	Short Stay	01 June 2025	Solo	14	722654	High Value
02 Checked-Out	None	83	31 August 2025	D1057	Short Stay	01 November 2025	Corporate	15	591073	High Value
29 Checked-Out	None	90	31 July 2025	D1033	Short Stay	01 October 2025	Solo	19	1136388	High Value
98 Checked-Out	None	48	06 October 2024	D693	Short Stay	01 November 2024	Corporate	16	762824	High Value
39 Checked-Out	None	55	07 December 2024	D762	Short Stay	01 January 2025	Solo	12	838656	High Value
80 Checked-Out	None	38	30 September 2025	D1042	Short Stay	01 November 2025	Solo	15	1304624	High Value
45 Checked-Out	None	27	08 June 2025	D917	Short Stay	01 July 2025	Corporate	15	1006366	High Value
75 Checked-Out	None	20	20 August 2025	D983	Short Stay	01 September 2025	Solo	16	1125704	High Value
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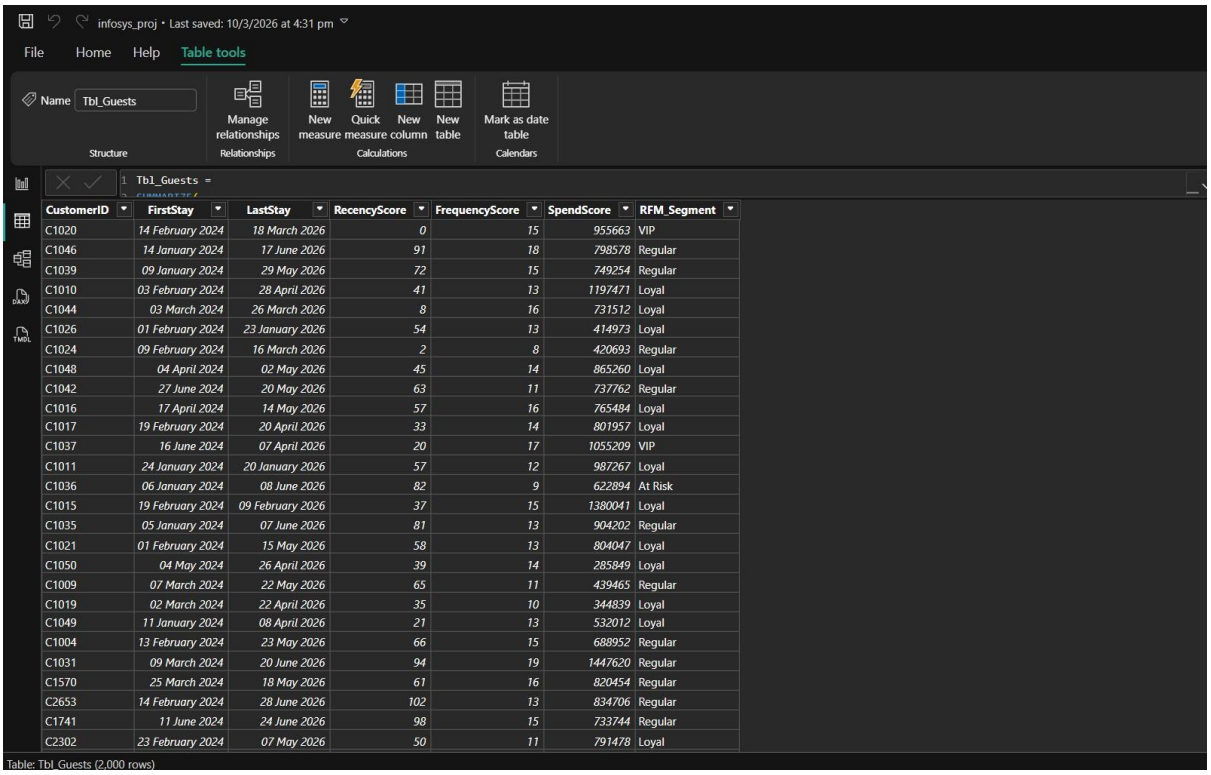
30 Checked-Out	None	6	08 March 2025	D804	Short Stay	01 March 2025	Solo	15	992285	High Value
14 Checked-Out	None	78	21 October 2023	D372	Short Stay	01 January 2024	Solo	9	554858	High Value
21 Checked-Out	None	37	17 November 2025	D1089	Short Stay	01 December 2025	Solo	8	441020	Regular
49 Checked-Out	None	79	27 August 2025	D1049	Short Stay	01 November 2025	Solo	10	229824	Regular

4. Data Transformation

A new table "Tbl_Guests" was created using aggregation functions to convert booking-level data into customer-level data.
Each row represents a unique customer with:

- FirstStay (earliest booking date)
- LastStay (most recent booking date)



CustomerID	FirstStay	LastStay	RecencyScore	FrequencyScore	SpendScore	RFM_Segment
C1020	14 February 2024	18 March 2026	0	15	955663	VIP
C1046	14 January 2024	17 June 2026	91	18	798578	Regular
C1039	09 January 2024	29 May 2026	72	15	749254	Regular
C1010	03 February 2024	28 April 2026	41	13	1197471	Loyal
C1044	03 March 2024	26 March 2026	8	16	731512	Loyal
C1026	01 February 2024	23 January 2026	54	13	414973	Loyal
C1024	09 February 2024	16 March 2026	2	8	420693	Regular
C1048	04 April 2024	02 May 2026	45	14	865260	Loyal
C1042	27 June 2024	20 May 2026	63	11	737762	Regular
C1016	17 April 2024	14 May 2026	57	16	765484	Loyal
C1017	19 February 2024	20 April 2026	33	14	801957	Loyal
C1037	16 June 2024	07 April 2026	20	17	1055209	VIP
C1011	24 January 2024	20 January 2026	57	12	987267	Loyal
C1036	06 January 2024	08 June 2026	82	9	622894	At Risk
C1015	19 February 2024	09 February 2026	37	15	1380041	Loyal
C1035	05 January 2024	07 June 2026	81	13	904202	Regular
C1021	01 February 2024	15 May 2026	58	13	804047	Loyal
C1050	04 May 2024	26 April 2026	39	14	285849	Loyal
C1009	07 March 2024	22 May 2026	65	11	439465	Regular
C1019	02 March 2024	22 April 2026	35	10	344839	Loyal
C1049	11 January 2024	08 April 2026	21	13	532012	Loyal
C1004	13 February 2024	23 May 2026	66	15	688952	Regular
C1031	09 March 2024	20 June 2026	94	19	1447620	Regular
C1570	25 March 2024	18 May 2026	61	16	820454	Regular
C2653	14 February 2024	28 June 2026	102	13	834706	Regular
C1741	11 June 2024	24 June 2026	98	15	733744	Regular
C2302	23 February 2024	07 May 2026	50	11	791478	Loyal

5. RFM Analysis

Recency
Recency measures how recently a customer has stayed. It is calculated as the difference between the last stay date and the current date.

Frequency
Frequency represents the total number of bookings made by each customer.

Monetary (SpendScore)
Monetary value represents the total revenue generated by each customer.

	RecencyScore	FrequencyScore	SpendScore
6	0	15	955663
6	91	18	798578
6	72	15	749254
6	41	13	1197471
6	8	16	731512
6	54	13	414973
6	2	8	420693
6	45	14	865260
6	63	11	737762
6	57	16	765484
6	33	14	801957
6	20	17	1055209
6	57	12	987267
6	82	9	622894
6	37	15	1380041
6	81	13	904202
6	58	13	804047
6	39	14	285849
6	65	11	439465
6	35	10	344839
6	21	13	532012
6	66	15	688952
6	94	19	1447620
6	61	16	820454
6	102	13	834706
6	98	15	733744
6	50	11	791478

6. RFM Segmentation

Based on RFM scores, customers were segmented into different categories:

- VIP: High-value, frequent, and recent customers
- Loyal: Frequent customers
- At Risk: Customers who have not visited recently
- Regular: Average customers

The thresholds were defined based on observed data distribution and business assumptions.

During analysis, it was observed that some date values resulted in negative recency due to future dates in the dataset. This was handled by adjusting the calculation to ensure meaningful results.

	RFM_Segment
3	VIP
8	Regular
4	Regular
1	Loyal
2	Loyal
3	Loyal
3	Regular
0	Loyal
2	Regular
4	Loyal
7	Loyal
9	VIP
7	Loyal
4	At Risk
1	Loyal
2	Regular
7	Loyal
9	Loyal
5	Regular
9	Loyal
2	Loyal
2	Regular
0	Regular
4	Regular
6	Regular
4	Regular
8	Loyal

7. Key Insights

- High-value customers (VIP) contribute significantly to total revenue
- Loyal customers show consistent booking behavior
- Some customers fall under the "At Risk" category, indicating potential churn (signs that a customer might stop coming back)
- Customer segmentation helps in targeted marketing and personalized services

Businesses can target At Risk customers with promotional offers to reduce churn. VIP customers can be rewarded with loyalty programs to increase retention.

This analysis can be further used to build dashboards for customer targeting and revenue optimization.